

Kai Ma

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ORCID

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Education

ShanghaiTech University

M.S. in Electronic Science and Technology

Focus: Physical design; Design for manufacturing

Supervisor: Dr. Hao Geng

China

Sep. 2024 – Present

Central China Normal University

B.S. in Electronic Science and Technology

Focus: Single-photon detector with low time jitter; Neural network optimization

China

Sep. 2020 – Jun. 2024

Publications

- [C1] **Kai Ma**, Zhen Wang, Hongquan He, Qi Xu, Tinghuan Chen, Hao Geng, “LMM-IR: Large-Scale Netlist-Aware Multimodal Framework for Static IR-Drop Prediction,” *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, Jun. 2025.
- [C2] **Kai Ma**, Tianyi Li, Jiaqi Liu, Jingyi Yu, Hao Geng, “No Pixel, More Efficient: Multimodal Framework for Sub-nm Mask Process Correction,” *IEEE/ACM Design, Automation and Test in Europe (DATE)*, Verona, Italy, Apr. 2026.
- [J1] Tianyi Li, **Kai Ma**(co-first author), Yiwen Wu, Jiaqi Liu, Jingyi Yu, Hao Geng, “Shot Splatting: Pioneering Shot-aware MPC for VSB Mask Data Preparation,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, doi: 10.1109/TCAD.2026.3667094.
- [C3] Wenxuan Dong, Tianyi Li, **Kai Ma**(co-first author), Zhen Wang, Hanbin Dong, Nan Wang, Su Zheng, Xinyun Zhang, Jingyi Yu, Hao Geng, “UniMRC: A Differentiable Multimodal Framework for Unified Curvilinear-Manhattan Mask Rule Checking,” *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, San Jose, Nov. 2026.
- [C4] Shenshuo Yao, Liuke Wang, Yuyang Chen, YINUO Zhu, Zhen Wang, Jiajie Zhang, Tianyi Li, Yiwen Wu, Xinheng Li, **Kai Ma**, Nan Wang, Wenxuan Dong, Xinyun Zhang, Jingyi Yu, Hao Geng, “Ariadne’s Thread: A Multimodal Benchmark Weaving Point Solutions into Process Chains for AI-Driven Computational Lithography,” *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, San Jose, Nov. 2026.

Patent

Single-photon detector for visible and near-infrared light in the short wavelength range with high count rate. CN 216668998 U. Han Xue et al., **Kai Ma**, Qinglin Wu.

Open-Source Repositories

TorchLitho — Contributor to an open-source differentiable computational lithography framework (**226 stars**); implements the first open-source differentiable Abbe / Hopkins lithography imaging model and analytical resist development model, enabling GPU-accelerated, gradient-based optimisation of Resolution Enhancement Techniques (RETs). (*SPIE Advanced Lithography*, 2024)

Research Experience

Multimodal EDA: IR-Drop Prediction & Mask Process Correction

Supervisor: Dr. Hao Geng

Sep. 2023 – Present

ShanghaiTech University, China

- Proposed **LMM-IR**, a multimodal framework for static IR-drop prediction that encodes large-scale SPICE netlists (>100K nodes) as 3D point clouds via a novel Large-scale Netlist Transformer (LNT), then fuses them with circuit image features through cross-attention; achieved the best F1 score and lowest MAE among ICCAD 2023 contest winners. (*DAC 2025*)
- Developed a multimodal **Mask Process Correction (MPC)** framework that represents GDS mask layouts as sparse point clouds and conditions correction on e-beam lithography resist images via a ControlNet-inspired architecture; achieved sub-nm EPE accuracy comparable to commercial tools at >200× speedup on large-scale layouts. (*DATE 2026*)
- Proposed **Shot Splatting**, a unified MPC and shot-fracturing framework; jointly optimises shot geometry, position, size, and dose via differentiable Lamé-curve kernels and EBL simulation, eliminating separate MRC post-processing and reducing EPE by ~79–86% over the no-MPC baseline. (*TCAD*)
- Proposed **OmniShot**, a shot-centric generative framework for Mask-Wafer Co-Optimization (MWCO) that unifies OPC, MPC, and shot fracturing into a single inference step; developed a CUDA-accelerated soft rasterizer and differentiable EBL & optical lithography simulators for physics-aware end-to-end training; achieved >40% lower EPE, >40% fewer shots, and 5× faster runtime over strong baselines on complex benchmarks and a full-chip DNN accelerator.

Low Time Jitter Single-Photon Detector

Sep. 2021 – Mar. 2023

Supervisor: Prof. Qinglin Wu

Central China Normal University, China

- Developed a novel constant fraction timing discrimination method to optimise single-photon detection time jitter, achieving a **50%** reduction.
- Designed an innovative signal processing circuit and conducted end-to-end performance validation.

Awards & Certifications

Sep. 2024	Graduate Academic Scholarship, ShanghaiTech University
Jul. 2024	Digital Implementation & Sign-off Full-Flow Certification, Cadence Design Systems — delivered by <i>Miao Liu</i> , Sr. Group Director.
Nov. 2022	National Encouragement Scholarship, Ministry of Education, China — <i>top 5% of undergraduates nationally</i>
Nov. 2021	National Encouragement Scholarship, Ministry of Education, China — <i>top 5% of undergraduates nationally</i>
Dec. 2021	Chinese Mathematics Competitions (CMC) — Second Prize

Technical Skills

Programming	Python · C/C++ · MATLAB · $\text{\LaTeX}$ · Linux
Expertise	Deep Learning · Virtuoso · Numerical Analysis

Conference Attendance

DAC 2025	ACM/IEEE Design Automation Conference, San Francisco, USA, Jun. 2025(Talk & Poster)
DATE 2026	IEEE/ACM Design, Automation and Test in Europe, Verona, Italy, Apr. 2026(Talk & Poster)
IWAPS 2025	International Workshop on Advanced Patterning Solutions, Shenzhen, China, 2025. Attended sessions on advanced lithography and mask patterning technologies.